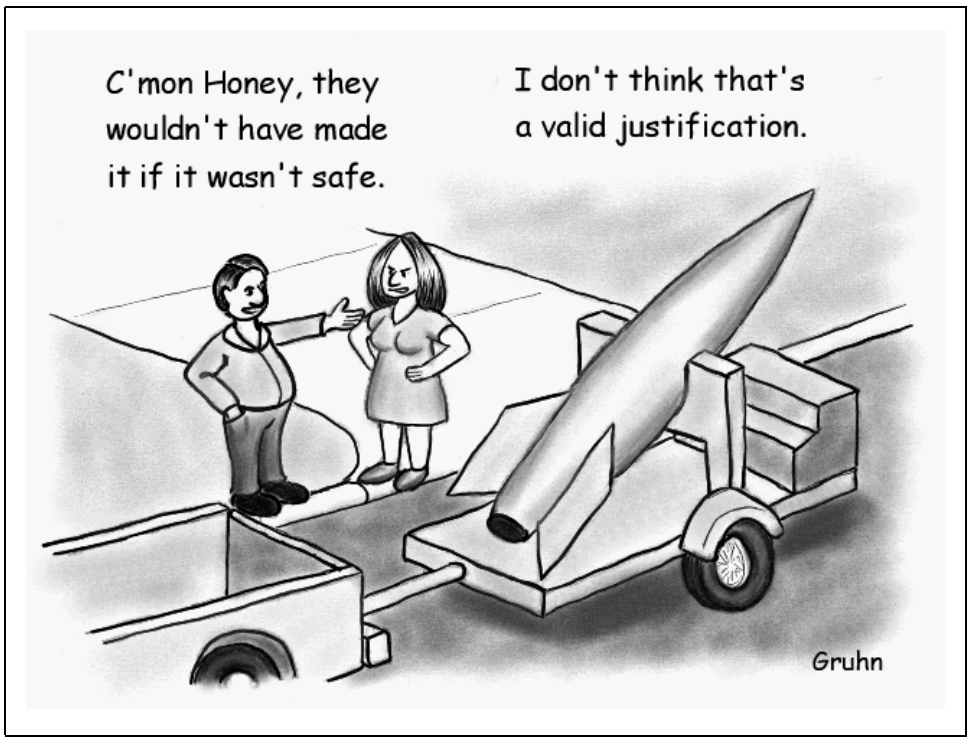
14

JUSTIFICATION FOR A SAFETY SYSTEM

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14.1 Introduction

In order to justify the need for a safety instrumented system (SIS), it's important to understand *why* such systems are installed in the first place. Safety systems are used in the process industries to lower the risk to personnel, the environment, loss of production, capital equipment and company image. In fact, it's not unusual to find that the safety integrity level (SIL) determined for some safety instrumented functions (SIF) are higher for economic or environmental reasons than for personnel protection.

The SIL determined for each SIF is usually based on practices developed by a company that follow industry standards (as covered in Chapter 6). The internal practices would normally be calibrated to reflect the amount of risk the company is willing to tolerate. The need to further justify a system may not be required as the justification process is an integral part of their internal practices.

Many companies install safety systems to address only the personnel protection issues and comply with regulatory requirements. In these instances, the SIL determined only reflects the "personnel SIL." With this approach, the assumption is that shutting down the process will address the safety concerns. There usually aren't any other needs for justification—the safety requirements have to be satisfied for legal, moral, financial, and ethical reasons. Justifying a safety system from a purely personnel safety point of view is usually not an issue.

Some companies prefer to have an additional justification step in order to demonstrate that the cost of providing the SIS is justified. If a detailed cost/benefit analysis is completed, the cost of providing the SIS may be shown to be too high and not justifiable compared to not installing the system at all. Justification for providing the system is, therefore, based on carefully analyzing the impact of each of the consequences.

In justifying the need for a safety system, one needs to look at two separate issues:

1. Can the safety instrumented functions be justified based on integrity levels selected to address risk receptors other than personnel (e.g., loss of production, equipment damage, etc.)?
2. The connection between the safety, reliability, and lifecycle costs of the safety system. The safety issue addresses the safety performance of the system based on the risk reduction required. In other words, will the system shut down when required? The reliability issue addresses the nuisance trip performance and the impact nui-

sance trips have on safety and overall costs. In other words, what impact will nuisance trips have on lost production costs and safety (due to the additional risks associated with unplanned shutdown and startup)? Lifecycle cost analysis addresses the total costs of the safety system over its entire life.

Therefore, a safety system can be justified not only for the safety integrity levels selected, but also based on its reliability and overall lifecycle costs.

Another issue facing control system designers is the use of safety instrumented systems, not in addition to, but *instead* of other non-SIS protection layers. Control system designers are sometimes requested to analyze and recommend the possible installation of safety systems to replace other layers because of cost, scheduling, or competitive reasons. A typical example is the installation of high integrity pressure protection systems (HIPPS) to eliminate the need for mechanical relief devices. Some plants have gone through enough upgrades and additions that the original flare systems may no longer be sized to accommodate a total plant-wide upset. It may be less expensive to install pressure protective instrumentation (so relief valves don't lift, thus potentially over-pressuring the flare system) than to change the flare system itself.

This chapter will address the following issues and their impact on safety system justification:

- Failure modes of safety systems
- Justification
- Responsibilities for justification
- How to justify
- Lifecycle cost analysis
- Optimizing safety, reliability, and lifecycle costs

14.2 Safety System Failure Modes

It's been stressed throughout this book that safety systems generally fail in two distinct modes (i.e., safe or dangerous). The impact of these failures on safety system justification can be better understood by further categorizing these two failure modes into four distinct types:

1. *Dangerous undetected failure:* The system has failed dangerously and will not respond to a demand, yet there is no indication of a problem.

2. *Dangerous detected failure:* The system has failed dangerously but the failure *is* annunciated to the operator through the internal self-diagnostics of the system. The system will not respond to a demand until the problem is rectified.
3. *Degraded mode due to a partial failure of a redundant system:* This applies to redundant systems that are fault tolerant. The failure of one channel (safe or dangerous) is annunciated to the operator through the internal self-diagnostics of the system, yet the system *will* respond to a demand.
4. *Nuisance trip failure:* A failure that produces a nuisance or spurious trip (i.e., an unplanned shutdown). The plant has to be restarted once the problem is corrected.

Mode 1: Dangerous Undetected Failure

This is usually considered the worst scenario. The probability of a dangerous undetected failure is the basis for the probability of failure on demand (PFD_{avg}) calculation and the safety integrity level (SIL) of the function or system (as covered in Chapter 8). The SIL determination should be based on the risk tolerance philosophy of the company. This policy should be well established and understood by all parties involved. If this is the case, the SIL determination is unlikely to be challenged by management for personnel protection issues. Whatever safety system architecture or level of redundancy is required in order to comply with the SIL determination is usually readily accepted. The only way to determine whether a dangerous undetected failure has occurred is through manual testing.

Mode 2: Dangerous Detected Failure

This would be the preferred mode if a dangerous failure were to occur. If the mean time to repair (MTTR) of a system was eight hours, and if a demand were to occur on average every six months, then the probability of a hazard occurring within the eight hour repair period would be approximately one in five hundred fifty (1/550).

Upon identifying such a failure, the operator would be expected to follow established procedures. Operating procedures may require the operator to:

1. Monitor the shutdown parameters and the overall process closer and be prepared to take manual action to bring the process to a safe state upon abnormal conditions. In extreme cases, this might require an immediate manual shutdown of the process.
2. Request assistance by support personnel to have the problem rectified. If the problem is not corrected within a predefined time limit,

the system may be designed to shut down automatically. This predefined time limit is usually designated by approval agencies (e.g., TÜV).

3. The level of diagnostic capabilities of the system—including field devices—needs to be established and analyzed as part of the overall system design. There may be additional costs associated with incorporating the level of diagnostics required. This needs to be justified.

Mode 3: Degraded Mode

Redundant systems have the capability to be fault tolerant (i.e., tolerate failures and continue to operate properly) and are widely used whenever both safety and availability/reliability issues need to be addressed. This is obviously the preferred failure mode, since the system will still function while the problem is being rectified. Redundancy increases the initial installation cost, but the overall lifecycle cost of the system may actually be lower. Sections 14.7 and 14.8 show one such example.

Mode 4: Nuisance Trip Failure

Standards such as IEC 61508 and 61511 focus on dangerous failures and performance issues (i.e., will the system shut down when needed) and do not fully address issues associated with safe failures (i.e., nuisance trips). Simply shutting a process unit down whenever there is a system fault is not always the safest or most economical thing to do. The following paragraph in 29 CFR 1910.119 summarizes the concerns about nuisance trips:

"In refining processes, there are occasionally instances when a piece of equipment exceeds what is deemed "acceptable," and interim measures are taken to bring the equipment back into conformance with safe operating parameters. Under (j)(4) it would be mandatory to immediately shut down the entire process upon discovery of such a situation. Shutdowns and startups are inherently dangerous operations, which we try to avoid unless absolutely necessary. In addition, the life expectancy of certain components is directly affected by the number of cycles to which they are subjected. We feel that safety is promoted rather than diminished by keeping shutdowns to a minimum."

It's not unheard of to find portions of safety systems disabled or bypassed when an unacceptable level of nuisance trips have a negative effect on production. Accidents have been reported by OSHA and the EPA as a result of such bypasses. [Ref. 3] To reduce the number of spurious trips, the system may require additional levels of redundancy (i.e., fault-tolerance).

This impacts the cost and complexity of the system and needs to be justified.

14.3 Justification

As mentioned in the introduction of this chapter, there are a number of risk receptors that may benefit from providing safety systems (i.e., personnel, the environment, loss of production, capital equipment and company image). Allocating monies to offset loss of life or injury to personnel can be a very contentious issue. Many companies avoid publishing or using such figures as part of a safety system justification. However, it is possible to quantify the financial impact of not providing a safety system for other reasons. Equipment damage often causes loss of production and these two factors may be combined. Some issues, such as the impact to company image, may be difficult to quantify, but may still be dealt with in more qualitative ways.

The justification for a safety system usually requires the expertise of control system engineering personnel since they have the cost data for safety system installation and operation. Justifications can be based on a cost-benefit analysis. The costs include the engineering, purchasing, installation, operation, and maintenance of a system. The benefits are the cost savings associated with reducing the number of injuries, incidents, and lost production. This chapter will focus on the justification from the following aspects:

1. equipment damage and associated production losses,
2. avoidance of spurious trips.

14.4 Responsibilities for Justification

The responsibilities for justifying a system may be different depending upon whether the system is being considered for protecting equipment and associated production losses, or for the avoidance of spurious trips. (As mentioned previously, justifying a safety system from a purely personnel safety point of view is usually not a problem.) For equipment damage and associated production losses, process and production engineering personnel may need to provide a lead role in the justification.

Issues often arise when recommendations are made to enhance or improve the reliability of a system in order to satisfy the nuisance trip requirements. This often increases the cost and/or complexity of the system. Justifications for such enhancements often face opposition for the following reasons:

- There is a perception by some that this is a production issue, not a safety issue. This is often a false perception.
- The objective of the safety system is to protect the plant. Unwarranted shutdowns due to failures of the safety system may be regarded as a control system issue that should be solved by control system personnel. Production and operations managers do not want to see the plant down for unnecessary reasons and may respond with, "What do you *mean*, your safety system will shut my plant down at least *once every year?!*"
- Project budgets are normally fixed at this stage. There may be reluctance to spend extra dollars unless it can be shown to be well justified.
- The data used for calculating the nuisance trip rate may be challenged.

Justifying a safety system that's above and beyond the SIL requirements is usually the responsibility of the control systems engineer. While the benefits of a more reliable safety system may appear to be obvious, don't assume that everyone will be in agreement. Some common questions will be:

- "The last project didn't require such a complex safety system. Why are you insisting on such complexity and extra cost now?"
- "Can we upgrade the system later if we really have a problem?"
- "What other less costly options have you considered? What's the best alternative?"

There is a need to properly justify the expenditure for any system. All projects are competing for fewer and fewer dollars. The impact of the expenditure on the bottom line will be closely scrutinized.

14.5 How to Justify

The justification for any system is usually based on a financial analysis. Therefore, a review of certain basic financial terms and functions, as well as how the value of money varies with time, is in order.

The future value (FV) of money varies with time and interest rate. If one were to make an annual fixed investment M, the future value of the investment after N years, at an interest rate R, can be expressed as:

$$FV = M\,\frac{[1+R]^N - 1}{R} \tag{14.1}$$

One can also calculate the present value (PV) of investments made at fixed intervals in the future. If one were to make an annual fixed investment M for N years, at an interest rate R, the present value of the investment can be expressed as:

$$PV = M\,\frac{1-[1+R]^{-N}}{R} \tag{14.2}$$

When justifying a safety system, we're interested in the present value of money based on losses that can be quantified on an annual basis over a number of years. In other words, one can calculate the impact of a hazardous event and/or nuisance trips on an annual basis, calculate the present value of future losses, and determine the limiting expenditure for the safety system. If the cost is greater than the benefits, the justification is questionable.

Justification Example # 1 - Equipment Damage and Associated Production Losses

A hazardous event may result in an explosion that will cause an estimated $1,000,000 in equipment damage. An analysis showed that the expected frequency of this event is 2.0×10^{-3}/year (1/500 per year). A proposed safety system with a risk reduction factor of 100 will reduce the overall risk to 2.0×10^{-5}/year (1/50,000 per year). Assuming an expected 20-year life of the safety system and an interest rate of 6%, what level of initial expenditure can be justified for the system?

Solution:

If the safety system reduces the risk from 2.0×10^{-3} /year to 2.0×10^{-5} / year then, on average, the annual savings will be $2000 - $20 per year, or $1980 ($1,000,000/500 - $1,000,000/50,000). The highest amount of money that can be justified for a safety system will be the present value of an investment that makes 20 annual payments of $1980 at an interest rate of 6%. This is calculated using equation 14.2:

$$PV = \$1{,}980 * [1 - (1.06)^{-20}]/0.06$$

$$PV = \$22{,}710$$

The above solution assumes the operation and maintenance costs of the safety system are negligible. In reality, these should also be factored into the calculations.

Justification Example # 2 - Avoidance of Spurious Trips

The total cost of a spurious trip for a unit is estimated at $100,000. The spurious trip rate of the existing safety system is once every 5 years. The safety system can be upgraded at a cost of $70,000 to decrease the spurious trip rate to once every 10 years. If the life of the system is assumed to be 20 years and the interest rate is 5%, is this a justifiable expenditure?

Solution:

The annual cost of spurious trip without an upgrade is $100,000/5 years, or $20,000 per year. The annual cost of spurious trip with the upgrade is $100,000 / 10 years, or $10,000 per year. The potential benefit is $10,000 per year ($20,000 - $10,000).

The annual cost of borrowing $70,000 over a period of 20 years at an interest rate of 5% is obtained by solving equation 14.2 for M.

$$= (\$70{,}000 * .05) \; / \; [1-(1+.05)^{-20}]$$

$$= \$5{,}617/\text{year}$$

The expenditure is justifiable since the benefit is greater than the cost.

14.6 Lifecycle Costs

One way to justify a safety system expenditure is to complete a lifecycle cost analysis for the various options being considered. The lifecycle costs reflect the total cost of owning the system. By calculating the lifecycle costs, various options can be analyzed in a more quantitative and consistent manner. Table 14-1 describes the predominant costs incurred during the life of a safety system. The list is divided into initial fixed costs (i.e., the costs for designing, purchasing, installing, commissioning, and operating the system), and annual costs (i.e., maintenance and other ongoing costs associated with the system). To some extent, the costs reflect the items listed in the lifecycle model covered in Chapter 2.

Table 14-1: Breakdown of Safety System Costs

Cost Item	Comments
Initial fixed costs	
SIL determination	The costs to complete the safety integrity level (SIL) determination (once the safety review is completed and a safety system is deemed necessary).

Safety requirements and design specifications	The manpower costs to complete the safety requirements specifications, conceptual design, and detailed design specifications.
Detailed design and engineering	Cost for the complete detailed design and engineering.
Sensors	Purchase cost of sensors.
Final elements	Purchase cost of valves and other final elements.
Logic system	Purchase cost of logic system.
Miscellaneous: power, wiring, junction boxes, operators interface.	Costs for other miscellaneous equipment required to install and monitor the safety system.
Initial training	Cost of training for design, operation, and support personnel to design, install, and test system.
FAT/Installation/PSAT	Costs for factory acceptance tests, equipment installation, and pre-startup acceptance tests.
Startup and correction	Most systems require some correction prior to full operation.
Annual costs	
Ongoing training	Ongoing refresher training for operations and support personnel.
Engineering Changes	These costs may be significant due to review requirements and documentation updates.
Service agreement	Programmable logic systems usually required a service agreement with the manufacturer in order to resolve “difficult” problems.
Fixed operation and maintenance costs	Utilities, preventive maintenance programs, etc.
Spares	Critical spares as recommended by vendors.
Online testing	Periodic testing carried out by operations and support personnel.
Repair costs	Costs for repairing or replacing defective modules based on predicted failure rates.
Hazard costs	Cost based on the hazard analysis. The hazard rate is a function of the PFD of the system and the demand rate.
Spurious trips costs	Cost of lost production based on the $MTBF^{spurious}$ of the system.
Present value for annual costs	The present value of the annual costs based on current interest rates and the predicted life of system. These costs are added to the initial fixed costs to obtain the present value of all costs. The PV may be calculated by solving the following equation: $$PV = M\,\frac{1-[1+R]^{-N}}{R}$$ where M is the annual cost, R is the interest rate and N is the number of years.
Total lifecycle costs	Total costs for the life of the system. This is the sum of the initial fixed costs and the present value for the annual fixed costs.

14.7 Review Example

The following example will be used to develop the lifecycle costs for two possible safety system solutions.

Flammable materials A and B are automatically and continuously fed in a fixed ratio to a reactor vessel by the basic process control system (BPCS). The setpoint of primary flow controller 1 is set by the vessel level controller in order to maintain a fixed level in the vessel. The flow controller for feed A adjusts the setpoint of the flow controller for feed B in order to maintain the fixed ratio.

Figure 14-1 shows the basic process controls.

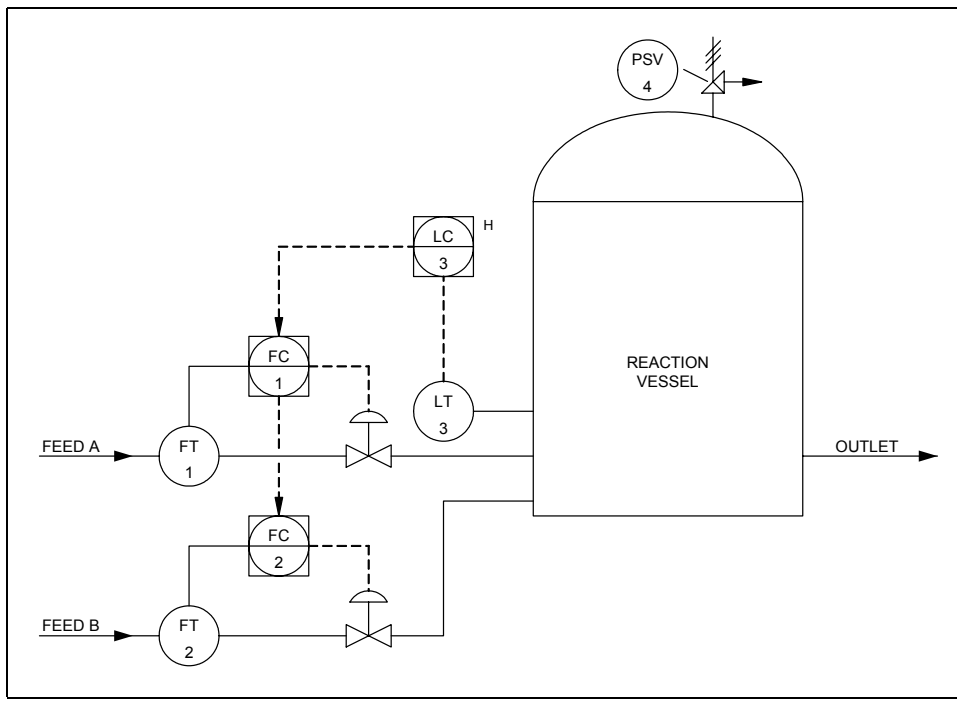


Figure 14-1: Basic Process Control for a Vessel

The following hazardous events were identified in a safety review:

Table 14-2: Summary of Hazard Analysis

Hazard	Cause(s)	Consequence	Likelihood of Occurrence
Release of flammable gas to environment	Failure of BPCS	Fire, explosion $500K loss	Medium
Vessel failure	Failure of BPCS and relief valve	$750K loss	Low
Note: A nuisance trip costs $10,000			

The following safety instrumentation was recommended.

1. Install a high pressure shutdown to close off feeds A and B to the vessel.
2. Install a high level shutdown to close off feeds A and B to the vessel.

Based on the above data, and using the 3-D SIL determination matrix described in Section 6.8 and Figure 6-2, SIL 1 is required for each safety function (pressure and level).

The following safety systems were proposed:

Case 1:

Sensors: Single transmitters

Logic: Relay logic

Valves: Single independent shutdown valves on each line

Case 2:

Sensors: Triplicated transmitters voted 2oo3 (mid value selection)

Logic: Fault-tolerant safety PLC

Valves: Single independent shutdown valves on each line

PFD_{avg} and $MTTF^{spurious}$ Calculations for Case 1

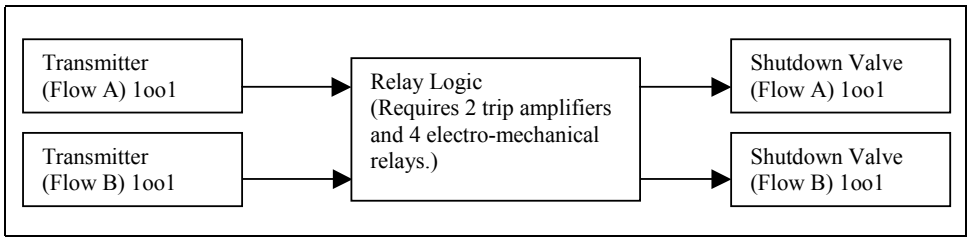


Figure 14-2: Proposed Safety System for Case 1

Failure data: (Failures/year)

Transmitter:	λ_{du} = 0.01	(100 yr MTTF)	λ_s = 0.02	(50 yr MTTF)
Valve and solenoid:	λ_{du} = 0.02	(50 yr MTTF)	λ_s = 0.1	(10 yr MTTF)
Trip amplifier:	λ_{du} = 0.01	(100 yr MTTF)	λ_s = 0.01	(100 yr MTTF)
Mechanical relay:	λ_{du} = 0.002	(500 yr MTTF)	λ_s = 0.02	(50 yr MTTF)

Additional data:

Test interval (TI): 6 months

PFD_{avg} Calculation

The SIL is determined for each individual function. Therefore, the probability of failure on demand (PFD) model should only include one function: 1 transmitter, 1 trip amplifier, 3 mechanical relays and both valves.

PFD_{avg}	=	λ_{du} * TI/2	
PFD_{avg} (Sensor)	=	0.01 * 0.5/2	= 0.0025
PFD_{avg} (Trip amplifier)	=	0.01 * 0.5/2	= 0.0025
PFD_{avg} (Mechanical relay)	=	3 * 0.002 * 0.5/2	= 0.0015
PFD_{avg} (Valve and solenoid)	=	2 * 0.02 * 0.5/2	= 0.0100
PFD_{avg} (Total)			**= 0.0165**

The maximum value allowed for a SIL 1 function is 0.1. Therefore, the above system satisfies the safety requirements. (The risk reduction factor [1/PFD] is 60, which is between 10 and 100 required for SIL 1.)

$MTTF^{spurious}$ Calculation

The mean time to failure, spurious ($MTTF^{spurious}$) calculation should include all components that may cause a shutdown: both transmitters, both trip amplifiers, 4 mechanical relays, and both valves with solenoids. Power supplies are assumed to be redundant and are ignored in the calculation (as non-redundant components will dominate).

$MTTF^{spurious}$	= $1/(\lambda_s)$		
$MTTF^{spurious}$ (Sensors)	= $1/(2 * 0.02)$	=	25 years
$MTTF^{spurious}$ (Trip amplifier)	= $1/(2 * 0.01)$	=	50 years
$MTTF^{spurious}$ (Mechanical relay)	= $1/(4 * 0.02)$	=	12.5 years
$MTTF^{spurious}$ (Valve and solenoid)	=$1/(2 * 0.1)$	=	5 years
$MTTF^{spurious}$ (Total)			**3 years**

A nuisance trip may be expected to occur, on average, every 3 years.

While the case 1 system meets the SIL 1 specification requirement, a system with redundant sensors and programmable logic will also be considered.

PFD_{avg} and $MTTF^{spurious}$ Calculations for Case 2

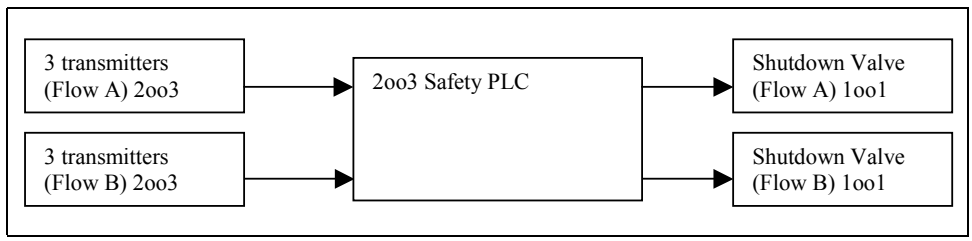


Figure 14-3: Proposed Safety System for Case 2

The failure rate data for the field devices will be the same as in Case 1. The logic system vendor usually provides the safe and dangerous performance of their hardware. Ten percent common cause is included for the redundant transmitters and logic system. Ninety-nine percent diagnostics are assumed for the redundant transmitter arrangement. Refer to Chapter 8 for the various formulas. Testing of the 2oo3 sensors and logic system will be done every 3 years. The valves will still be tested every 6 months.

PFD_{avg} Calculation

The SIL is determined for each individual function. Therefore, the probability of failure on demand (PFD) model should only include one function: 2oo3 transmitters, 2oo3 safety PLC, and both valves. Only the common cause portion of the redundant components needs to be calculated (as it will dominate).

PFD_{avg} (2oo3 sensors)	=	0.01 * 0.01 * 0.1 * 3/2	= 0.000015
(failure rate * undetected failure % * common cause % * test interval/2)			
PFD_{avg} (2oo3 safety PLC)	=	(from vendor)	= 0.00005
PFD_{avg} (Valve and solenoid)	=	2 * 0.02 * 0.5/2	= 0.0100
PFD_{avg} (Total)			**= 0.01**

The maximum value allowed for an SIL 1 function is 0.1. Therefore, this system also satisfies the safety requirements. (The risk reduction factor [1/PFD] is 100, which is the borderline between SIL 1 and SIL 2.)

MTTF^{spurious} Calculation

The mean time to failure, spurious (MTTF^{spurious}) calculation should include all components that may cause a shutdown: both 2oo3 transmitter arrangements, 2oo3 safety PLC, and both valves with solenoids. Only the common cause portion of the redundant components needs to be calculated (as it will dominate).

MTTF^{spurious} (2oo3 Sensors) = 1/(2 * 0.02 * 0.1)	=	250 years
(1 / quantity * failure rate * common cause %)		
MTTF^{spurious} (Safety PLC) = (from vendor)	=	200 years
MTTF^{spurious} (Valve and solenoid)=1/(2 * 0.1)	=	5 years
MTTF^{spurious} (Total)	=	**5 years**

A nuisance trip may be expected to occur, on average, every 5 years. The valves represent the weak link in Case 2 (both for safety and nuisance trips), as the other portions of the system are fault tolerant. The case 2 system is safer than case 1 and offers fewer nuisance trips. The following lifecycle cost analysis further compares the two systems.

14.8 Detailed Lifecycle Cost Analysis

This analysis is intended to demonstrate the technique for completing the lifecycle cost analysis. *The data is very subjective* and is *not* intended to reflect the actual costs for any particular product or installation technique. Individuals will have to rely on data within their own organization for more accurate numbers.

Table 14-3: Lifecycle Costs for Case 1

	Material $	Labor $	Total cost $	Subtotal $
Initial Fixed Costs				
SIL determination		$2,000	$2,000	
SRS/Design specifications		$5,000	$5,000	
Detailed design and engineering		$30,000	$30,000	
Sensors	$4,000		$4,000	
Final elements	$3,000		$3,000	
Logic system	$10,000		$10,000	
Miscellaneous: power, wiring, junction boxes	$5,000		$5,000	
Initial training		$5,000	$5,000	
FAT/Installation/PSAT	$5,000	$25,000	$30,000	
Startup and correction	$2,000	$8,000	$10,000	
Fixed costs subtotal				$104,000
Annual costs				
Ongoing training		$3,000	$3,000	
Engineering changes	$1,000	$4,000	$5,000	
Service agreement				
Fixed operation and maintenance costs		$1,000	$1,000	
Spares	$2,000		$2,000	
Manual testing		$25,000	$25,000	
Repair costs	$1,000	$500	$1,500	
Hazard costs			$12,000	
Spurious trip costs			$4,000	
Annual costs subtotal				$45,500
PV for annual costs				$567,031
(20 year life, 5% interest rate)				
Total lifecycle costs				**$671,031**

Table 14-4: Lifecycle Costs for Case 2

	Material $	Labor $	Total cost $	Subtotal $
Initial fixed costs				
SIL determination		$2,000	$2,000	
SRS/Design specifications		$5,000	$5,000	
Detailed design and engineering		$30,000	$30,000	
Sensors	$12,000		$12,000	
Final elements	$3,000		$3,000	
Logic system	$30,000		$30,000	
Miscellaneous: power, wiring, junction boxes	$5,000		$5,000	
Initial training		$15,000	$15,000	
FAT/Installation/PSAT	$5,000	$25,000	$30,000	
Startup and correction	$2,000	$8,000	$10,000	
Fixed costs subtotal				$142,000
Annual costs				
Ongoing training		$3,000	$3,000	
Engineering changes	$1,000	$1,000	$2,000	
Service agreement		$2,000	$2,000	
Fixed operation and maintenance costs		$1,000	$1,000	
Spares	$4,000		$4,000	
Manual testing		$15,000	$15,000	
Repair costs	$1,000	$500	$1,500	
Hazard costs			$6,000	
Spurious trip costs			$2,000	
Annual costs subtotal				$36,500
PV for annual costs				$454,871
(20 year life, 5% interest rate)				
Total lifecycle costs				**$596,871**

The Case 2 system has fewer nuisance trips and improved safety compared to Case 1. In addition to the lower manual testing costs (for less frequent testing of the sensors and logic system), the overall lifecycle costs are lower, meaning the Case 2 system is preferred and can be justified.

14.9 Optimizing Safety, Reliability, and Lifecycle Costs

Many chose to focus only on the logic box in the past. Yet simple modeling will usually show that field devices have a much more significant impact, not only on safety performance, but nuisance trip performance and overall lifecycle costs as well. Various options (e.g., redundancy, test intervals, levels of automatic diagnostics, common cause, etc.) can be analyzed and compared based on their long term impact on safety, reliability, and cost.

Reliability modeling and lifecycle cost analysis are very effective tools for optimizing the design of a safety system. Modeling and lifecycle cost analysis allows one to focus on the specific elements of the safety system that require increased attention. By means of the models and the cost analysis, one can better justify system configurations and test intervals based on long term benefits. As shown in the example above, a fault tolerant system may initially cost more but may have lower lifecycle costs due to fewer nuisance trips and reduced testing costs.

This approach can also be used to analyze whether a safety instrumented system can be used as an alternative to other independent protection layers. It's essential that process engineers, as well as other system specialists be involved in the overall analysis.

Summary

Most companies have practices in place to identify the need for safety systems to address personnel safety issues. In such cases, justifying a safety system is usually not an issue.

However, safety systems may be justified not only for personnel safety reasons, but for reliability and total lifecycle cost benefits as well. Other factors that may justify the use of a safety system include protection to capital equipment, loss of production, the environment, and company image. Reliability modeling and lifecycle cost analysis are ideal tools to justify the design and installation of safety systems.

References

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